

LINGUAFORCE: Benchmarking Agentive Force of Discourse in Dialogues via Unified Measurement Dimensions

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Abstract—Language can push a listener toward an action, narrow the available choices, or do both. This paper presents LINGUAFORCE, a benchmark for measuring this negative agentive force in de-identified multi-turn dialogues. It contains 4,066 dialogues in a 3,432-dialogue training split and a disjoint 634-dialogue held-out split. Each dialogue is represented with three linked components: a taxonomy of four families and fifteen types, seven measurement dimensions, and linguistic cue guidelines for interpreting each score. The benchmark evaluates label consistency, agreement between two independent annotators, three prediction tasks, a two-stage dimension-aware pipeline, and transfer to MentalManip, MultiManip, TalkDown, and ToxicChat.

Index Terms—Agentive force, computational social science, psycholinguistic framework, large language models, manipulation detection.

I. Introduction

Language can change what a listener sees as possible or acceptable. A request, threat, moral appeal, or deceptive statement can make one action easier to choose and other actions harder to choose. This listener-directed effect is called agentive force of discourse. The benchmark focuses on its negative forms: language that pressures, constrains, or steers a listener against a neutral choice. Ordinary requests, advice, and rational persuasion are included as controls.

Existing NLP resources cover parts of this problem. Toxicity and condescension datasets focus on explicit hostility [1], [2]; persuasion datasets often target news text and rhetorical framing [3]; and dialogue datasets on psychological manipulation focus on narrower covert strategies such as gaslighting and intimidation [4]–[6]. Compliance-gaining research offers useful strategy taxonomies [7]–[9], but these resources use different label spaces. The central question is whether different negative language strategies can be represented and measured in one space. Transfer to strategies outside the annotation is a downstream test of that representation.

To address this gap, LINGUAFORCE defines a shared representation and tests whether it is useful for prediction and transfer. Its contributions are:

- 1) A unified three-layer framework. The framework combines a four-family, fifteen-type strategy taxonomy with seven measurement dimensions and cue

guidelines. The dimensions separate the force of a push, its effect on the listener’s options, its pressure channel, and its directness.

- 2) A new annotated benchmark. The release contains 4,066 dialogues with dialogue-level intensity and seven-dimension annotations, organized into a training split and a disjoint held-out split. The benchmark relates dimension scores to binary pressure and intensity labels.
- 3) An experimental protocol. The benchmark defines detection, strategy recognition, and intensity prediction tasks. It also provides a two-stage dimension-aware pipeline, a direct zero-shot baseline, and tests of dimension structure and transfer to unseen types.

II. Related Work

A. Datasets of Manipulation and Persuasion

Compliance-gaining research provides the conceptual starting point. Marwell and Schmitt [7] and Wiseman and Schenck-Hamlin [8] proposed strategy inventories, and Cialdini [9] organized common influence principles such as reciprocity, social proof, and authority. In NLP, SemEval-2023 Task 3 detects persuasion techniques in online news [3]. MentalManip, MultiManip [4], [5], and ReaMent [6] study covert manipulation in dialogue. These resources use different units and label spaces. Many omit ordinary non-coercive controls such as requests, advice, and rational persuasion. Toxicity [1], condescension [2], and microaggression datasets [10] cover hostile language, but hostility is only one possible pressure channel.

LINGUAFORCE addresses this gap by putting ordinary non-coercive controls and negative, listener-restricting strategies in one dialogue-level measurement space. It does not replace these resources; it provides a common representation for comparing the pressure they study.

B. Speech Acts and Pressure

Speech-act theory studies how people use language to perform actions such as requesting, ordering, and promising [11], [12]. This perspective helps explain why aggressive words alone cannot identify pressure. For example, “A good daughter would stay” contains no insult, but invokes a family obligation to pressure the listener to stay. The

analysis identifies the action requested by the speaker and the ways the dialogue makes refusal harder. The annotation records pressure expressed in the dialogue, not whether the listener actually complies.

C. Normative Influence

Normative influence encourages compliance by appealing to duties, social roles, moral expectations, or authority. For example, “You are the team leader; it is your duty” uses a role obligation to pressure the listener. Research on moral foundations provides background for understanding moral and authority-based appeals [13]. These forms of influence motivate the normative-pressure dimension in our framework.

III. Framework: A Unified Account of agentive force of discourse

Negative agentive force is represented at three levels. The strategy layer describes what the speaker does. The measurement layer records how much pressure is applied. The cue layer links each judgment to words and discourse patterns that can support it.

A. Definition and Inclusion Criterion

The agentive force of discourse of an utterance or dialogue is the pressure it places on a listener’s behavior or decision space. It can push the listener toward the speaker’s preferred action or make other actions harder to choose. There are three properties of agentive force. First, the pressure does not need to succeed; it can be present even when the listener resists. Second, one utterance can use more than one channel, such as guilt and social proof. Third, pressure has a degree, so strategies can differ in strength as well as type.

A discourse has agentive force of discourse when it targets a recognizable listener action or decision and tries to drive or restrict that behavior. The pressure may come from a request, inducement, norm, emotion, deception, or hostility. The taxonomy provides labels for the cases covered in this dataset, but it is not intended to cover every possible strategy. Pure information exchange and emotional venting without a behavioral target are negative examples.

B. Three-Layer Architecture

Layer 1 describes the speaker’s tactic with fifteen types in four families. Layer 2 records the amount and source of pressure along seven dimensions. Layer 3 provides guidelines for connecting scores to observable words, constructions, and discourse patterns. For example, an ultimatum can receive a high option-constraint score because it removes alternatives, while an indirect moral appeal can receive a high normative-pressure score even without an insult. The three layers make different strategies comparable and provide evidence for error analysis.

C. Strategy Taxonomy

Table I presents the taxonomy. The four families organize the current negative-force scope by the dominant mechanism. Family A uses open requests, advice, orders, and reasoning. Family B motivates through an expected benefit or harm. Family C is social-normative: it asks the listener to act because of a duty, role, moral rule, authority, past favor, or group norm. Family D uses less direct tactics, including emotional exploitation, deception, false dilemmas, and hostility. These families are descriptive organizing units, not mutually exclusive psychological causes, and one dialogue may receive multiple type labels. The final column in Table I records the dominant mechanism used by each type; these mechanisms are descriptive and do not imply mutually exclusive causes.

D. Measurement Dimensions

Table II defines seven scores on a continuous 0–1 scale and gives each score a four-level label. Scores are assigned at both turn and dialogue level. The scores describe different parts of the same pressure. They are a measurement design for this benchmark, not a complete list of psychological traits. Their value depends on agreement, discrimination, and transfer.

The dimensions answer different questions. D_1 measures how strongly the speaker pushes for an action. D_2 measures how much the listener’s choice space is narrowed. A direct request can have high D_1 but low D_2 if the listener still has several clear alternatives. An ultimatum can have high D_2 because it presents few acceptable alternatives. D_3 , D_4 , and D_6 identify different pressure channels: social norms, emotional leverage, and overt hostility. A message may use one channel without using the others.

D_5 measures factual or memory distortion. For example, “I never said that; you misremember” can be deceptive even when it is polite. An insult can be hostile without being deceptive, so D_5 and D_6 are separate. D_7 measures how directly the pressure is expressed. “You must stay” is more explicit than “A good daughter would stay”, but explicitness alone does not determine pressure strength. Together, the seven scores describe the push, the effect on available choices, the pressure channel, and the form of the expression.

The data also support simple checks of the dimension design. A moral-pressure dialogue should usually have high D_3 , while a polite indirect appeal may have lower D_7 . Family-level profiles and clustering test whether different strategy families concentrate on different dimensions.

E. Linguistic Cue Guidelines

Table III links each dimension to typical language cues. The cues are evidence, not automatic triggers. For example, “must” may indicate directive force only when the dialogue uses it to push the listener toward an action. A cue can be ambiguous, absent, or dependent on earlier turns. The checklist therefore supports both

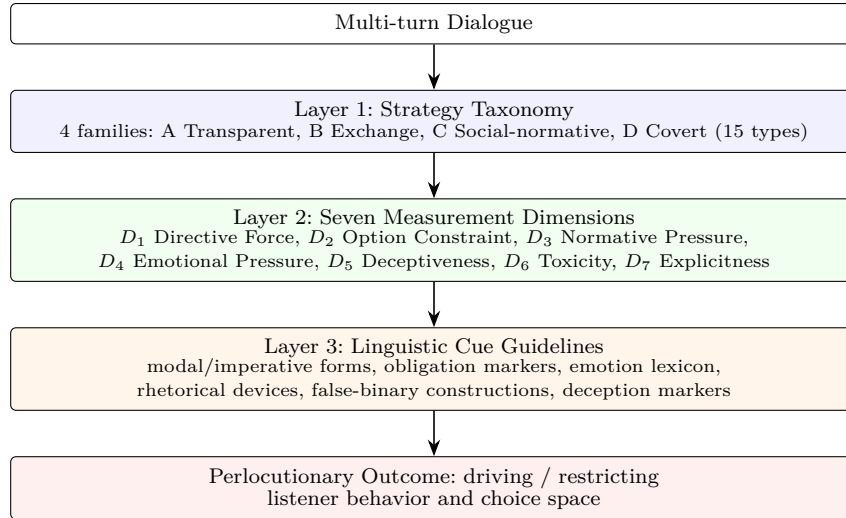


Fig. 1. Three-layer architecture of LINGUAFORCE. The strategy taxonomy defines what the speaker does, the measurement dimensions measure how much pressure is applied, and the cue guidelines link the scores to observable wording. Normative pressure is expected to be high when the speaker invokes duty, morality, or social roles, while explicitness depends on whether that pressure is stated directly or indirectly.

annotation and error analysis, and the resulting scores can be explained with the cues that support them.

F. A Worked Annotation Example

Table IV shows a worked annotation. The dialogue mixes moral appeal (C1) and obligation (C2), hence receives a multi-label type annotation; intensity is high and the profile is dominated by D_3 with secondary D_1 . The example illustrates why the task is hard: the language is polite (no D_6) and the pressure implicit (low D_7)—the pattern that defeats lexical toxicity classifiers.

Two short examples show why context matters. In a normal request, one speaker says, “Could you close the window?” and the other replies, “Sure.” This is an A1 request with little pressure. The same request can carry more pressure when it follows “I helped you last time,” because the added context invokes C6 reciprocity. The wording alone is almost unchanged, but the dialogue-level profile changes because the listener’s decision is framed as a debt.

G. Dimension Structure and Interdependence

The dimensions are not assumed independent; their correlation structure is an empirical object with explicit, falsifiable expectations. (i) D_1 should correlate with D_2 (the harder the push, the narrower the options), with transparent A1 as the predicted exception. (ii) D_3 and D_4 should co-occur in family C (guilt and moral appeal are emotionally loaded) but separate in family B (threats are affectively cold). (iii) D_5 should be roughly orthogonal to the others—gaslighting occurs in both polite and hostile registers—so a strong correlation with D_6 or D_7 would indicate annotation bias. (iv) D_7 should correlate negatively with D_5 : covert strategies are typically indirect. The statistical analysis (Section IV) tests these via correlation

matrices and per-family profiles; deviations are reported and used to revise the guideline rather than discarded.

IV. Dataset Construction and Analysis

A. Design Principles

The dataset must cover the intended negative-force scope, provide comparable labels for every dialogue, and include hard negatives that separate pressure from mere toxicity or ordinary requests. Four principles guide construction. (i) De-identified multi-turn dialogues: conversational interaction with the turn structure and pragmatic context needed for dialogue-level analysis. (ii) Scope coverage: negative-force and non-force dialogue across the 0–5 range, so benign controls prevent degenerate models that classify any request as manipulation. (iii) Comparable measurement: every instance receives the same seven-dimension annotation on one scale. (iv) Hard negatives: polite coercion, sarcastic requests, and superficially kind manipulation stress-test lexical shortcuts.

B. Sources and Sampling

The release contains 4,066 de-identified, everyday multi-turn dialogues. We use 3,432 dialogues as the training split and 634 disjoint dialogues as the held-out split (Section IV-F). Each dialogue has binary pressure and 0–5 intensity annotations, which are used to evaluate the seven-dimension profiles. The release contributes new annotations, a unified taxonomy, and an evaluation protocol. The dialogues are de-identified, and provenance and licensing information are documented with the release.

C. Annotation Protocol and Negative Cases

The annotation follows the decision tree in Figure 2. First, the annotator checks whether the dialogue targets a listener action or decision. If it does not, the dialogue

TABLE I

Strategy taxonomy of LINGUAFORCE: four families and fifteen types. The mechanism column records the dominant pressure channel for each type.

Family	Type	Description and example	Mechanism	Role
A. Transparent	A1 Request/Advice	Low-pressure, intention-transparent proposal. “Could you revise this report by Friday?”	—	benign control
	A2 Directive/Order	Direct instruction with expected compliance. “Send the report before you leave.”	—	transparent pressure
	A3 Rational Persuasion	Argument, evidence, expert opinion. “The data show 30% lower cost; adopt it.”	—	reasoned pressure
B. Exchange	B1 Promise/Inducement	Benefit-driven motivation. “Help me this once and I will return the favor.”	—	incentive pressure
	B2 Threat/Warning	Harm-driven motivation. “If you do not sign, you will face consequences.”	constraint	explicit constraint
	C1 Moral Appeal/Guilt	Appeals to conscience or moral duty. “A dutiful son would never let his mother suffer.”	normative	normative core
C. Social-normative	C2 Obligation/Duty	Invokes role responsibilities. “You are the team leader; it is your duty.”	normative	duty pressure
	C3 Value Judgment/Shaming	Moral labeling of the target. “You are so selfish; you never think of others.”	normative	moral attack
	C4 Authority	Appeals to status, rank, seniority. “The boss decided; why keep arguing?”	—	hierarchical pressure
D. Covert	C5 Conformity/Social Proof	Appeals to group norms. “Everyone is doing it; why are you special?”	—	peer pressure
	C6 Reciprocity/Debt	Invokes past favors. “I helped you last time; do not refuse this.”	—	debt pressure
	D1 Emotional Manipulation	Feigned weakness, pity, or fear to steer behavior. “If you leave too, I do not know what to do.”	—	affective leverage
	D2 Deception/Gaslighting	Distorts facts or memory. “I never said that; you misremember.” ¹	—	epistemic attack
	D3 Logical Trap/False Dilemma	Shrinks the option space. “Either you follow me or we are done.”	constraint	option elimination
	D4 Verbal Abuse/Toxicity	Explicit hostility and disparagement. “You are useless; you cannot even do this.”	hostility	hostile pressure

TABLE II

Seven measurement dimensions used to measure negative agentive force.

Dim.	Definition	Role
D_1 Directive Force	Strength with which the utterance drives a target action	direct pressure
D_2 Option Constraint	Degree to which the listener’s options are narrowed	choice restriction
D_3 Normative Pressure	Pressure via duty, morality, norms, roles	social steering
D_4 Emotional Pressure	Use of guilt, shame, fear, pity as leverage	emotional leverage
D_5 Deceptiveness	Degree of factual or epistemic distortion	epistemic attack
D_6 Toxicity	Explicit hostility and abuse	hostile pressure
D_7 Explicitness	Transparency vs. indirectness of the pressure	language form

is a negative example. Second, the annotator assigns all applicable strategy types. For example, one dialogue may contain both a moral appeal and social proof. Third, the annotator assigns the 0–5 intensity and the seven dimension scores. Each dimension receives a continuous 0–1 score and a four-level label: None, Low, Moderate, or High.

Negative examples use the same dimension scale as

TABLE III

Illustrative language cues per dimension. Cues are indicative, not sufficient: a cue only counts when the discourse targets the listener’s behavior.

Dim.	Typical cues
D_1	imperative and hortative forms; modal “must/should/have to”; performative requests
D_2	“either/or”, ultimatums, no-alternative framings, threats of consequence
D_3	duty and role nouns, moral adjectives, normative “ought”; appeal to what “a good X” would do
D_4	emotion lexicon (guilt, shame, fear, pity), emotional blackmail, self-deprecation
D_5	fact distortions, memory disputes, gaslighting patterns, manufactured consensus
D_6	insults, disparagement, profanity, dehumanizing labels
D_7	direct address and explicit request vs. hints, presuppositions, innuendo

positive examples. Typical negative examples include pure information exchange, emotional venting with no behavioral target, and information-seeking questions. Advice that also acts as a subtle request receives every applicable type, while the dimension scores describe its dominant pressure channel. The released annotations are produced by an instruction-tuned LLM at temperature 0 using this protocol.

D. Quality Control and Consistency Validation

Quality control combines three checks: agreement between independent annotators, consistency with the binary pressure and intensity labels, and the expected monotonicity of predicted intensity. For each dimension, the analysis reports Spearman correlations with the binary

TABLE IV

A worked annotation example. S = speaker turn; L = listener turn. Type, intensity, and the seven dimension scores are annotated per turn and aggregated at dialogue level.

Turn	Utterance	Type (multi-label)	Dimension profile
S1	“A dutiful son would never let his mother live alone; you are not that kind of person, are you?”	C1 Moral Appeal; C3 Value Judgement	$D_3=0.9$, $D_1=0.7$, $D_4=0.6$, $D_7=0.4$
L1	“I know, but my job requires me to move...”	— (defensive)	$D_1=0.2$
S2	“Your mother sacrificed everything for you; it is your turn to repay her.”	C2 Obligation; C6 Reciprocity	$D_3=0.9$, $D_2=0.5$, $D_1=0.8$
Dialogue-level			Intensity 4/5; T1 positive; C1+C2+C3+C6

TABLE V
Released LINGUAFORCE sets.

Set	Dialogues	Coercive	Non-coercive
Training split	3,432	1,862 (54.2%)	1,570 (45.8%)
Held-out split	634	332 (52.4%)	302 (47.6%)

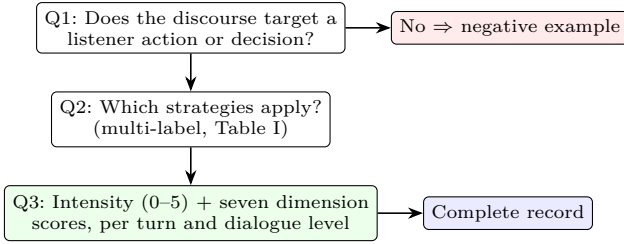


Fig. 2. Three-question annotation decision tree.

and 0–5 labels and its AUC for separating coercive from non-coercive dialogues (Table VIII). It also checks whether predicted intensity increases with the gold label and whether the inter-dimension correlation matrix contains pathological coupling (Figures 5 and 6). These checks test whether the dimensions provide reproducible structure beyond a single binary judgment.

Two graduate students in Computer Technology, including the first author, independently annotated 150 dialogues using a shared annotation rubric. Both annotators were proficient in reading English dialogues. Neither annotator had access to the model outputs during annotation. The sample was drawn from the 3,432-dialogue training split by randomly selecting 75 dialogues from each binary-label class without replacement, followed by shuffling (random seed 2026). This balanced design covers both classes; the reported agreement therefore describes this sample rather than an unweighted estimate for the full dataset. Table VI reports inter-annotator agreement on the discrete labels. Intensity reaches substantial agreement (quadratic-weighted $\kappa=0.823$), as do normative pressure (0.830), toxicity (0.820), option constraint (0.776), directive force (0.771), and emotional pressure (0.758). Explicitness reaches moderate agreement (0.543). These estimates concern discrete labels and do not directly establish the reliability of the continuous model scores.

TABLE VI

Inter-annotator agreement on a stratified subsample ($n=150$; two independent annotators, 150 fully labeled pairs). Binary uses unweighted Cohen’s κ ; intensity and dimensions use quadratic-weighted κ . Agreement levels use the cutoffs $\kappa \geq 0.60$ (substantial), $0.40 \leq \kappa < 0.60$ (moderate), and $\kappa < 0.40$ (low).

Variable	Cohen’s κ	Agreement level
binary (pressure present)	0.739	substantial
intensity (0–5)	0.823	substantial
D_1 directive force	0.771	substantial
D_2 option constraint	0.776	substantial
D_3 normative pressure	0.830	substantial
D_4 emotional pressure	0.758	substantial
D_5 deceptiveness	0.847	substantial
D_6 toxicity	0.820	substantial
D_7 explicitness	0.543	moderate

E. Statistical Analyses

This analysis tests whether the seven dimensions capture distinct parts of pressure. It compares each score with the binary and 0–5 labels, inspects inter-dimension correlations, compares profiles for coercive and non-coercive dialogues, and tests whether predicted intensity follows the ordered labels. Type-level results are reported in Section V-H.

F. Consistency Check on the Held-Out Set

The held-out analysis asks whether the seven-dimensional profile follows the pressure and intensity labels before the full set is analyzed. Agreement is a necessary check, but it does not show that the dimensions cause changes in listener behavior. The held-out split contains 634 dialogues annotated under the unified seven-dimension scheme. It contains 332 coercive (52.4%) and 302 non-coercive dialogues, and gold intensity is distributed as 170/54/78/64/107/161 over levels 0–5, so hard negatives and the high-intensity tail are both represented.

Table VII reports the consistency of the extracted profiles with the binary and intensity annotations. All seven dimensions correlate with the binary pressure label and with the 0–5 intensity label; the strongest channels are option constraint (D_2), directive force (D_1), and emotional pressure (D_4), followed by explicitness (D_7) and normative pressure (D_3); this is consistent with constraint being a core channel of moral-pressure labels. Each dimension separates coercive from non-coercive dialogues with AUC between 0.72 and 0.83. The predicted intensity rises

TABLE VII

Consistency of the seven dimensions with the binary and intensity annotations on the held-out split ($n=634$). ρ : Spearman correlation; AUC: area under the ROC curve for separating coercive from non-coercive dialogues.

Dimension	ρ (binary)	ρ (0-5)	AUC
D_1 Directive force	0.54	0.60	0.81
D_2 Option constraint	0.57	0.63	0.83
D_3 Normative pressure	0.49	0.63	0.78
D_4 Emotional pressure	0.54	0.60	0.81
D_5 Deceptiveness	0.49	0.55	0.77
D_6 Toxicity	0.42	0.35	0.72
D_7 Explicitness	0.53	0.53	0.79
Predicted intensity	—	0.66	0.83

TABLE VIII

Seven-dimension profiles on the training split ($n=3,432$). Mean score, Spearman correlation ρ with the binary label and the 0-5 intensity label, and AUC for separating coercive from non-coercive dialogues.

Dimension	Mean	ρ (binary)	ρ (0-5)	AUC
D_1 Directive force	0.53	0.51	0.54	0.79
D_2 Option constraint	0.36	0.55	0.58	0.82
D_3 Normative pressure	0.49	0.45	0.55	0.76
D_4 Emotional pressure	0.53	0.57	0.62	0.83
D_5 Deceptiveness	0.12	0.48	0.50	0.77
D_6 Toxicity	0.22	0.40	0.31	0.71
D_7 Explicitness	0.73	0.51	0.49	0.79
Predicted intensity	—	0.63	0.65	0.85

monotonically with the gold label (Spearman $\rho=0.66$), and the inferred binary decision reaches AUC 0.83 and F1 0.80 at the fixed intensity threshold of 3. These results suggest that the unified dimension space captures much of the signal in the binary measurement on this split. Section IV-G gives the corresponding descriptive statistics for the 3,432-dialogue training split.

G. Training-Split Descriptive Analysis

This subsection describes the label distribution and dimension profiles in the 3,432-dialogue training split. It is descriptive; independent evaluation is performed on the disjoint 634-dialogue held-out split. The training split contains 3,432 dialogues, including 1,862 coercive (54.2%) and 1,570 non-coercive (45.8%) dialogues. Gold intensity is distributed as 986/267/317/337/621/904 over levels 0-5. Level-0 hard negatives account for 28.7% of the data, while levels 4-5 account for 44.4%.

Table VIII summarizes the training-release profiles. Emotional pressure (D_4) and option constraint (D_2) have the strongest univariate separation (AUC 0.829 and 0.815), followed by directive force (D_1 , 0.792) and explicitness (D_7 , 0.788). Toxicity (D_6) is weaker (AUC 0.710) and therefore overlaps with, but does not define, the coercive core. No single dimension reaches the performance of the combined profile. The predicted intensity also increases with the gold 0-5 label (Spearman $\rho=0.65$), which is consistent with the intended ordinal interpretation.

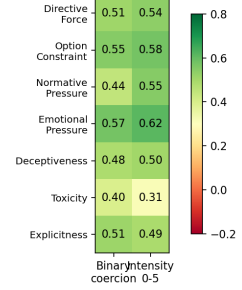
Seven dimensions vs. coercion labels, full set ($n=$ 

Fig. 3. Seven dimensions versus the binary and 0-5 intensity labels on the training split ($n=3,432$; Spearman correlation).

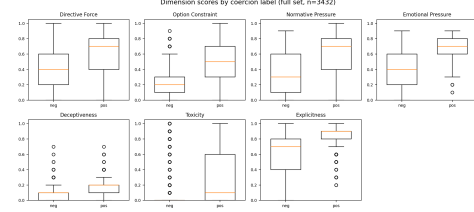


Fig. 4. Dimension scores for coercive versus non-coercive dialogues on the training split ($n=3,432$).

Ethics and data scope. The dataset contains de-identified English dialogues, and the release adds no new personal information or data collection. Detailed provenance, human-subjects considerations, annotation welfare, dual-use risk, cultural scope, and the accompanying datasheet are provided in Appendix A.

V. Experimental Design and Evaluation Protocol

The experiments test whether the agentive-force profile supports prediction beyond description. T1 measures pressure detection, T2 strategy recognition, and T3 intensity prediction. Ablation and cross-domain tests measure the role of individual dimensions and transfer to other resources.

A. Tasks

LINGUAFORCE defines three tasks at increasing granularity. T1 (agentive-force detection) is a binary classification of whether a dialogue exerts negative pressure beyond ordinary non-coercive controls. T2 (type recognition) is a fifteen-way multi-label classification of the strategies in Table I; type labels are released for both the held-out split and the training split. T3 (intensity prediction) is ordinal regression of the 0-5 intensity, evaluated with quadratic weighted kappa (QWK) and accuracy-within- ± 1 .

B. Two-Stage Pipeline

The two-stage pipeline tests whether the explicit profile adds useful structure beyond a direct judgment. An upstream LLM converts each dialogue into turn-level and dialogue-level dimension scores, and a downstream model uses them for T1-T3. For comparison, a direct judge predicts the label from the dialogue without producing the

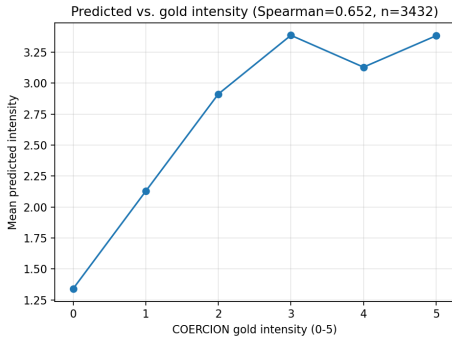


Fig. 5. Predicted intensity as a function of the gold 0–5 intensity label on the training split; points are within-level means.

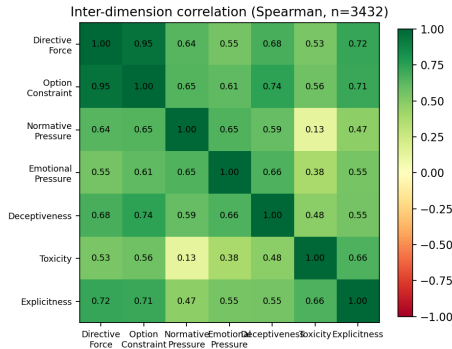


Fig. 6. Inter-dimension correlation matrix on the training split (Spearman).

seven-dimension profile. The experiments compare three ways to combine information: turn-only, dialogue-only, and turn-plus- dialogue representations.

The pipeline is evaluated zero-shot: a frozen instruction-tuned LLM parses each dialogue into the profile, and a direct judge that skips the profile serves as control. The main results use DeepSeek. A second frozen parser, GPT-6 Astra, is evaluated on the held-out consistency set to test parser sensitivity. Prompt variants include zero-shot, chain-of-thought [14], and self-reflection, following the self-perception approach of MultiManip [5] (Table IX).

C. Experimental Setup

All parsers run at temperature 0. T1 reports binary AUROC and F1 at the fixed intensity threshold of 3; T3 uses Spearman correlation, QWK, and accuracy within ± 1 . T2 reports macro- and micro-F1; its exploratory best-threshold values use the candidate thresholds described in Table XIII. Cross-domain transfer uses the same frozen parser on four public corpora.

Parser sensitivity is evaluated on the held-out split with the same dimensions, prompt format, temperature, and output schema; the corresponding results are reported with the main results.

TABLE IX
Model configuration for the zero-shot evaluation in this release.

Model	Role	Tuning
DeepSeek	upstream dimension parser	frozen
DeepSeek	direct zero-shot judge	frozen
GPT-6 Astra	upstream dimension parser	frozen (held-out check)

TABLE X
Aggregation ablation of the dimension-conditioned pipeline (linear readout; train on the training split, held-out split). T mean-pools the per-turn profiles over turns; B concatenates the turn-mean and global profiles.

Aggregation	Binary AUC	Intensity ρ	QWK
G (global only)	0.855	0.674	0.667
T (turn mean)	0.525	0.040	0.066
B (turn+global)	0.859	0.679	0.680

D. Research Questions

The experiments examine four questions: whether the dimensions agree with the pressure annotations, whether they provide complementary information, whether they transfer to related forms of pressure, and whether the results are robust to implementation choices. RQ1 (annotation consistency): Do the seven dimensions align with both the binary pressure and 0–5 intensity annotations? The evaluation reports each dimension’s Spearman correlation with these labels and its AUC for separating coercive from non-coercive dialogues. Results are reported separately for the disjoint held-out split ($n=634$) and the training split ($n=3,432$) in Sections IV–F–IV–G.

RQ2 (dimension structure): Do the seven dimensions carry distinct, complementary signal? Report per-dimension scores and AUCs, examine the inter-dimension correlation matrix for expected couplings (e.g., D_3 – D_4) and unexpected ones (e.g., D_5 – D_6), and analyze per-family profiles (Section V–H).

RQ3 (zero-shot generalization): Can the profile detect pressure in strategies that are absent from the training data? We apply the frozen pipeline to external corpora and evaluate leave-one-type-out transfer within the LINGUAFORCE taxonomy. The results are reported in the main results and cross-domain sections.

RQ4 (implementation choices): How do aggregation and prompt design affect the pipeline? We compare global-only, turn-mean, and combined aggregation with the same linear readout, and evaluate zero-shot, chain-of-thought, and self-reflection prompts. The results are reported in the main results and Tables X–XI.

E. Cross-Domain Evaluation

The cross-domain test asks whether the profile transfers to related listener-directed pressure while remaining distinct from toxicity without a behavioral target. The zero-shot pipeline is applied to MentalManip and MultiManip (psychological manipulation), TalkDown (patronizing lan-

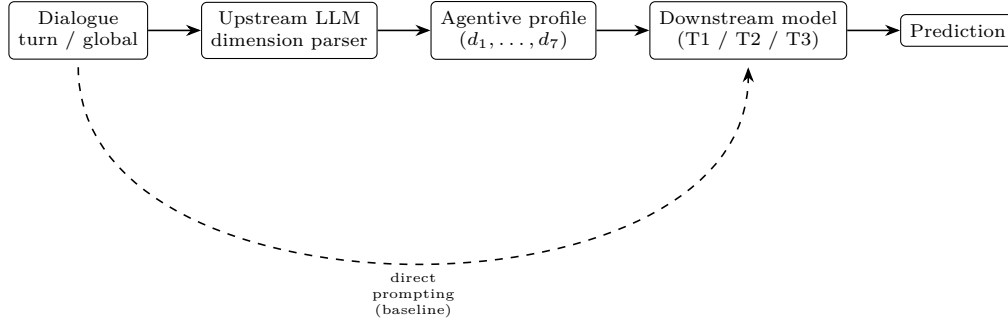


Fig. 7. Two-stage dimension-aware pipeline. The dashed path is the direct prompting baseline used to isolate the effect of dimension anchoring.

TABLE XI

Prompt ablation of the upstream dimension parser (held-out split, $n=634$). All variants use the same downstream linear readout.

Prompt variant	T1 AUC	T3 Spearman	T3 QWK
Zero-shot	0.832	0.666	0.623
Chain-of-thought	0.839	0.639	0.613
Self-reflection	0.838	0.670	0.630

guage), and ToxicChat (toxicity). Each corpus is evaluated with its standard metric and threshold-free AUROC.

F. Dimension Contribution and Ablation

This analysis asks whether the seven dimensions provide complementary information. Three checks are used. First, each dimension is compared with the gold pressure and intensity labels. Second, a simple linear model uses the seven scores to predict the labels. Third, each dimension is removed in turn to measure its marginal contribution.

The linear model, trained on the training split and evaluated on the held-out split, reaches binary AUC 0.855 and intensity Spearman 0.674. These results are close to, and slightly higher than, the profile-based LLM pipeline (AUC 0.831, Spearman 0.665). Both systems use the seven scores produced by the parser; the comparison therefore shows that a simple downstream model can recover the main predictive signal from the profile. It does not show that the linear model replaces the upstream LLM. The leave-one-dimension-out results in Table XII show different roles for different dimensions. Removing toxicity (D_6) hurts pressure detection most (AUC drop 0.008), while removing normative pressure (D_3) hurts intensity prediction most (Spearman drop 0.021). Although D_6 has the weakest individual correlation, it helps separate hostile dialogues that do not contain a behavioral target. D_4 overlaps substantially with D_3 ; removing D_4 slightly improves both metrics, which suggests redundancy or noise in this readout. The remaining dimensions have small marginal effects. These changes should be read as marginal effects, not as evidence that any single dimension is indispensable. Overall, the results support a combined representation rather than reliance on one decisive dimension.

TABLE XII

Leave-one-dimension-out ablation with a linear readout over the seven dimension scores (held-out split, $n=634$). ρ : Spearman correlation of predicted intensity with the gold 0–5 label. The LLM readout is the dimension-conditioned zero-shot reference.

Readout	Binary AUC	Intensity ρ
7-dim linear readout	0.855	0.674
– D_1 (directive force)	0.854	0.674
– D_2 (option constraint)	0.851	0.668
– D_3 (normative pressure)	0.851	0.653
– D_4 (emotional pressure)	0.860	0.693
– D_5 (deceptiveness)	0.855	0.674
– D_6 (toxicity)	0.847	0.669
– D_7 (explicitness)	0.853	0.674
LLM readout (reference)	0.831	0.665

G. Dimension Differences by Strategy Family

This section compares the seven-dimensional profiles of the strategy families. The goal is to see whether different families show different patterns across the dimensions, while allowing for overlap between related strategies. Figure 8 uses t-SNE only for visualization: it maps the 3,432 dialogue profiles to two dimensions and colors them by dominant family, with the 226 dialogues containing no detected strategy shown as benign. The families show some separation in this view, while benign dialogues tend to lie on the low-pressure side. To quantify family-level structure, we train a multinomial logistic classifier on the seven dimensions to predict the four strategy families plus the benign group. Five-fold cross-validation gives an accuracy of 0.640, compared with 0.200 chance accuracy. The profiles therefore contain family-level signal, although the moderate accuracy indicates substantial overlap between families. Each dimension alone separates benign from manipulative dialogues with AUC 0.71–0.83, supporting pressure detection but not family-specific separation. Figure 9 reports each family’s mean dimension profile: the social-normative family peaks on normative and emotional pressure (D_3 , D_4), the exchange family on option constraint and toxicity (D_2 , D_6), the covert family on emotional pressure (D_4), and benign dialogues are uniformly low across all dimensions.

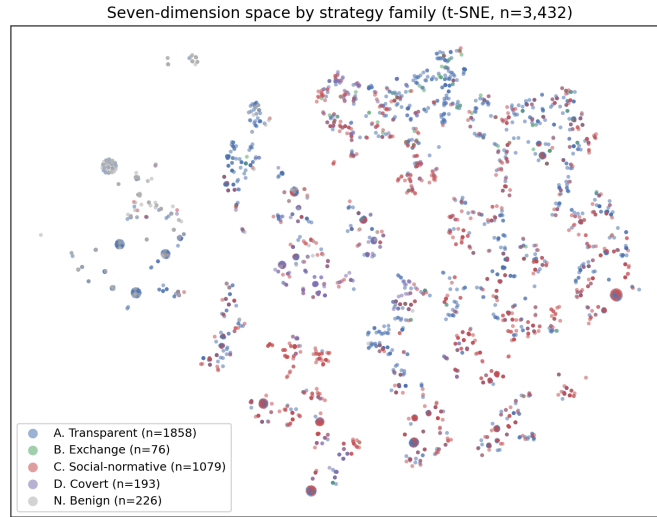


Fig. 8. t-SNE embedding of the seven-dimension profiles ($n=3,432$), colored by dominant strategy family; the 226 no-strategy dialogues form the benign group.

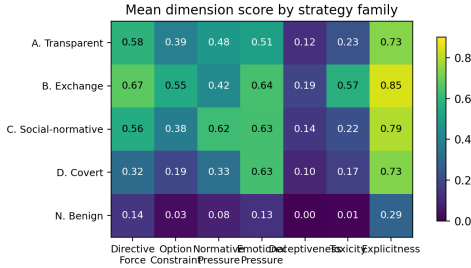


Fig. 9. Mean dimension score per strategy family. Social-normative and covert families concentrate pressure on normative/emotional dimensions; benign dialogues are uniformly low.

H. Main Results

Results are reported for pressure detection and intensity prediction, cross-domain transfer, strategy recognition, and implementation choices.

Pressure detection and intensity. Table XIV and Table XV report the dimension-conditioned pipeline on the 3,432-dialogue training split and the disjoint 634-dialogue held-out split. The pipeline first maps each dialogue to a seven-dimension profile (Section V) and then reads out the binary pressure label and the 0–5 intensity. On the training split, it reaches a binary AUC of 0.852 and an F1 of 0.817 at threshold 3; the corresponding held-out values are 0.831 and 0.795. Intensity prediction reaches Spearman $\rho = 0.651$, QWK 0.595, and accuracy within ± 1 of 0.692 on the training split, compared with $\rho = 0.665$ and QWK 0.623 on the held-out split. These values are similar across the two splits. On the held-out split, a direct judge that skips the profile reaches AUC 0.780 and F1 0.755 at threshold 3; this comparison does not isolate the contribution of individual pipeline steps. The held-out parser comparison gives AUC 0.831 for DeepSeek and 0.838 for GPT-6 Astra, with intensity Spearman 0.665

and 0.629, respectively. Binary detection is similar, while intensity ranking differs between parsers.

Cross-domain transfer. The cross-domain results in Table XVI evaluate the frozen pipeline on four external corpora. Threshold-free AUROC is 0.742 on MentalManip, 0.706 on MultiManip, and 0.729 on TalkDown, indicating moderate transfer to psychological manipulation and patronizing language. Transfer to ToxicChat is weaker (AUROC 0.587), consistent with generic toxicity being less tied to a listener-directed behavioral target. At the fixed source threshold of 3, F1 is much lower on TalkDown (0.058), so ranking performance does not imply that the same decision threshold transfers. The table therefore reports AUROC as the primary cross-domain metric and fixed-threshold scores only as descriptive reference values.

Strategy recognition. Table XIII compares a one-vs-rest logistic readout over the seven dimensions with a supervised RoBERTa-base classifier. Training on the 3,432-dialogue training split and evaluating on the held-out split gives the dimension readout family-level macro/micro-F1 of 0.772/0.827 and type-level macro/micro-F1 of 0.480/0.668. Five-fold cross-validation on the training release gives similar values (family 0.752/0.820; type 0.477/0.669). RoBERTa scores 0.780/0.820 at the family level and 0.544/0.731 at the type level on the held-out split. The dimension readout is competitive at the family level, while RoBERTa performs better on individual types. Three types have fewer than 30 training examples: rational persuasion (A_3), authority (C_4), and deception/gaslighting (D_2); their scarcity warrants caution when interpreting type-level macro-F1. The reported F1 thresholds were selected on the respective evaluation sets, so these are exploratory best-threshold scores rather than performance at a prespecified threshold.

The leave-one-type-out analysis yields mean novelty

TABLE XIII

T2 type recognition: a one-vs-rest logistic readout over the seven dimension scores versus fine-tuned RoBERTa-base. “Held-out”: trained on the training split ($n=3,432$), evaluated on the type-labeled held-out split ($n=634$); “5-fold CV”: cross-validation over the training split. F1 is reported as macro/micro; thresholds are selected on each evaluation set from $\{0.3, 0.4, 0.5, 0.6\}$, so these are best-threshold estimates.

Granularity	Protocol	Linear readout		RoBERTa-base
		Macro-F1	Micro-F1	Macro/Micro-F1
15 types	Held-out	0.480	0.668	0.544 / 0.731
15 types	5-fold CV	0.477	0.669	0.540 / 0.734
4 families	Held-out	0.772	0.827	0.780 / 0.820
4 families	5-fold CV	0.752	0.820	0.767 / 0.815

TABLE XIV

T1 detection on the training split ($n=3,432$): the dimension-conditioned readout, versus the same readout on the held-out split ($n=634$) and a direct zero-shot judge that skips the profile (dashed path in Fig. 7). F1 uses the fixed intensity threshold of 3.

Metric	Training ($n=3,432$)	Held-out ($n=634$)	Direct zero-shot
Binary AUC	0.852	0.831	0.780
F1 @ intensity ≥ 3	0.817	0.795	0.755
Accuracy @ intensity ≥ 3	0.789	0.760	0.711
Best per-dimension AUC (D_2)	0.829	0.825	—

AUROC 0.54, close to chance. Separately, predicted intensity distinguishes dialogues with any annotated strategy from those with none (AUROC 0.963). This latter test includes known strategies and does not measure detection of held-out types on its own.

Implementation choices. The implementation ablations in Tables X–XI show that turn-only mean pooling is near chance (AUC 0.525), whereas combining turn-level and dialogue-level profiles gives the best result (AUC 0.859, Spearman 0.679, QWK 0.680). The three prompt variants yield similar results (AUC 0.83–0.84), indicating limited sensitivity to prompt wording.

I. Limitations and Threats to Validity

The evidence has four boundaries. First, the seven dimensions are a measurement design rather than a complete theory of influence, and the 150-dialogue agreement study is a limited reliability check. Second, the release contains English dialogues from sampled everyday scenarios; transfer results on four external corpora do not establish cross-cultural or cross-lingual validity. Third, parser comparisons cover only two models, and F1 values whose thresholds are selected on the evaluation set are exploratory rather than fixed-threshold estimates. Finally, the labels measure pressure expressed in dialogue, not whether a listener complied or was harmed.

J. Discussion

The experiments support a practical interpretation of the framework. A single profile makes heterogeneous negative strategies comparable, while the cross-domain

TABLE XV

T3 intensity prediction from the dimension-conditioned pipeline on the training split ($n=3,432$), with the held-out ($n=634$) values for comparison.

Metric	Training	Held-out
Spearman ρ	0.651	0.665
Pearson r	0.677	0.691
Quadratic weighted kappa	0.595	0.623
Accuracy within ± 1	0.692	0.722
Exact match	0.216	0.219

TABLE XVI

Cross-domain zero-shot transfer of the seven-dimension pipeline. “Standard metric” is each corpus’s own metric under the source-task decision rule (intensity ≥ 3); “AUROC” is threshold-free and directly comparable. Sampled subsets: MentalManip $n=300$, MultiManip $n=220$, TalkDown $n=652$, ToxicChat $n=300$.

Corpus	Standard metric	Zero-shot	AUROC
MentalManip	macro-F1	0.615	0.742
MultiManip	macro-F1	0.347	0.706
TalkDown	F1	0.058	0.729
ToxicChat	AUROC	0.587	0.587

results suggest that some of their signal transfers beyond the training taxonomy. The profile also makes predictions inspectable through dimension-level evidence. These findings support measurement and analysis; they do not imply that the dimensions are independent psychological traits or that the benchmark predicts individual responses.

VI. Conclusion

This paper presents LINGUAFORCE, a dataset and benchmark for measuring negative agentive force in multi-turn dialogue. The release provides intensity labels, seven dimension profiles, strategy labels, and an evaluation protocol. The experiments show that the profiles support pressure detection, intensity ordering, and transfer to related manipulation and patronizing-language corpora. The benchmark is a measurement and analysis resource, not a causal model of listener behavior. Future work should broaden the annotation study, evaluate more parsers, and test other languages and forms of influence.

Appendix A Ethics and Privacy

The dialogues are de-identified English text. The release adds no personal information or new data collection. The study does not collect behavioral or demographic data; the manual checks use already-anonymized text. Annotators may skip distressing content and take breaks. Because the taxonomy could be misused to craft manipulative language, the release is intended for research on detection and analysis, not automated moderation, surveillance, or consequential decisions without human review. The current data are English and drawn from everyday scenarios, so cultural and cross-lingual validity remains an open question. A datasheet documents composition, preprocessing, intended uses, and limitations.

References

- [1] Z. Lin, Z. Wang, Y. Tong, Y. Wang, Y. Guo, Y. Wang, and J. Shang, “Toxicchat: Unveiling hidden challenges of toxicity detection in real-world user-ai conversation,” in Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, 2023, pp. 4694–4702.
- [2] Z. Wang and C. Potts, “Talkdown: A corpus for condescension detection in context,” in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, 2019, pp. 3711–3719.
- [3] J. Piskorski, N. Stefanovitch, G. D. S. Martino, and P. Nakov, “Semeval-2023 task 3: Detecting the category, the framing, and the persuasion techniques in online news in a multi-lingual setup,” in Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023), Toronto, Canada, July 2023, pp. 2343–2361.
- [4] Y. Wang, I. Yang, S. Hassanpour, and S. Vosoughi, “Mentalmanip: A dataset for fine-grained analysis of mental manipulation in conversations,” in Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Bangkok, Thailand, 2024, pp. 3747–3764.
- [5] D. Khanna, P. Seth, S. Murali, A. Guru, S. Shukla, T. Tyagi, S. Chaurasia, and K. Ghosh, “Self-percept: Introspection improves large language models’ detection of multi-person mental manipulation in conversations,” in Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), Vienna, Austria, 2025, pp. 660–675.
- [6] Y. Gao, P. Gao, H. Bao, B. Li, J. Luo, Z. Wang, and W. Chen, “Boosting large language models for mental manipulation detection via data augmentation and distillation,” arXiv preprint arXiv:2505.15255, 2025.
- [7] G. Marwell and D. R. Schmitt, “Dimensions of compliance-gaining behavior: An empirical analysis,” *Sociometry*, vol. 30, no. 4, pp. 350–364, 1967.
- [8] R. L. Wiseman and W. J. Schenck-Hamlin, “A multidimensional scaling validation of an inductively-derived set of compliance-gaining strategies,” *Communication Monographs*, vol. 48, no. 4, pp. 251–270, 1981.
- [9] R. B. Cialdini, *Influence: The Psychology of Persuasion*. William Morrow, 1984.
- [10] L. Breitfeller, E. Ahn, D. Jurgens, and Y. Tsvetkov, “Finding microaggressions in the wild: A case for locating elusive phenomena in social media posts,” in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), Hong Kong, China, 2019, pp. 1664–1674.
- [11] J. L. Austin, *How to Do Things with Words*. Oxford University Press, 1962.
- [12] J. R. Searle, *Speech Acts: An Essay in the Philosophy of Language*. Cambridge University Press, 1969.
- [13] J. Graham, J. Haidt, and B. A. Nosek, “Liberals and conservatives rely on different sets of moral foundations,” *Journal of Personality and Social Psychology*, vol. 96, no. 5, pp. 1029–1046, 2009.
- [14] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, “Chain-of-thought prompting elicits reasoning in large language models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.